

Annexure No.:

EXPERIMENT - 13

Aim: Demonstrate the working of PIR sensor

Apparatus:

Arduino uno, PIR sensor, LED, Resistor, Breadboard, wires, USB cable.

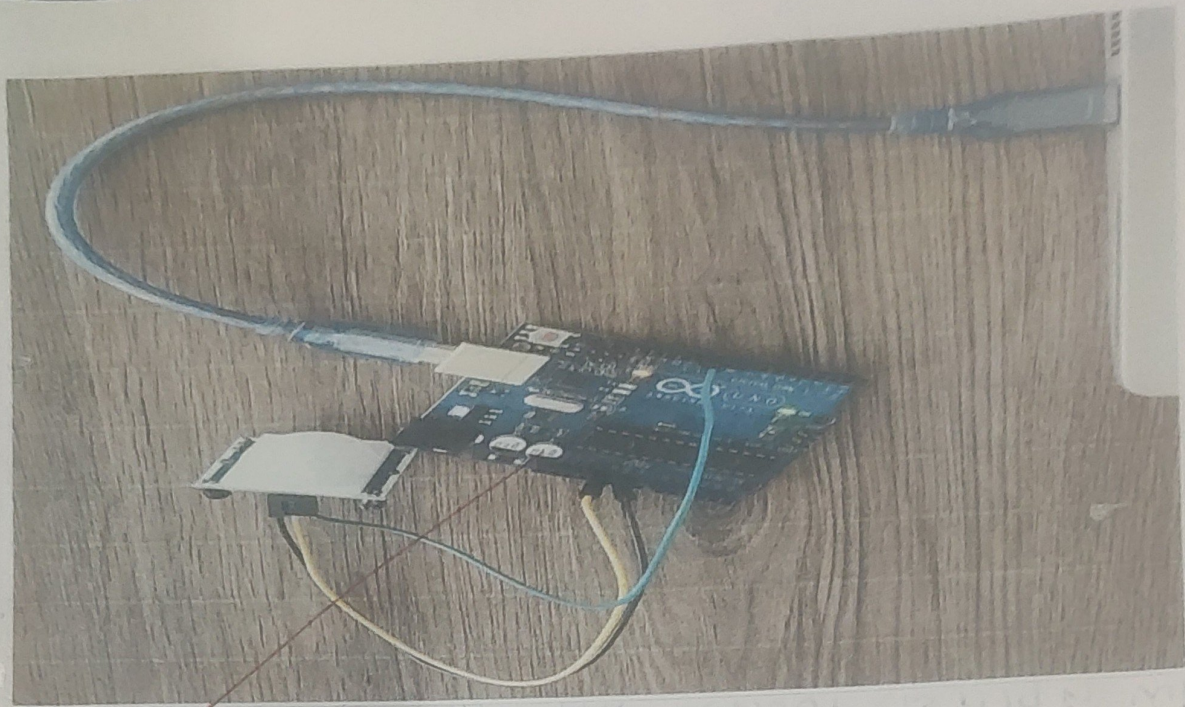
Theory:

PIR sensor is used for detecting infrared heat radiations. This makes them useful in applications involving detection of moving living objects that emit infrared heat radiations.

The output of PIR sensor is high when it sensor is high when it sensor motion; Whereas it is low when there is no motion.

- PIRs are basically made of a pyroelectric sensor, which can detect levels of infrared radiation.
- Above figure of PIR element shows the round metal can with a rectangular crystal in the center.

Performance Result:



The output of IIR sensor is shown in the figure below. It shows the output of the sensor is a square wave. This makes it clear that the IIR sensor is used for detecting motion.

Conclusion:

Breadboard, wires, USB cable.

Arduino Uno, IIR sensor, LED, Resistor.

Apparatus:

This demonstrates the working of IIR sensor.

EXPERIMENT-12

Annexure No.:

CODE :

```

Sketch: pir.ino
1 int pir = 8; // PIR sensor pin
2
3 void setup() {
4   pinMode(pir, INPUT);
5   Serial.begin(9600); // to see output on Serial monitor
6 }
7
8 void loop() {
9   if (digitalRead(pir) == HIGH) {
10    Serial.println("Motion Detected");
11   } else {
12    Serial.println("No Motion");
13   }
14   delay(1000);
15 }
  
```

Output: Serial Monitor X

Motion Detected
Motion Detected
Motion Detected
Motion Detected
Motion Detected
No Motion
No Motion

Upload error: Failed uploading: uploading error: exit status 1

COPY ERROR MESSAGES

conclusion:

The PIR sensor with Arduino UNO effectively detects motion and enables automatic control systems, making it useful security and automation application.

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